

GRAPH SIGNAL PROCESSING OF BRAIN CONNECTOME NETWORKS: SPECTRAL ANALYSIS AND GEOMETRIC DEEP LEARNING FOR GLIOBLASTOMA PREDICTION

ADITYAKRISHNA SREERAMACHANDRARAO, ANUSHA LAKSHMI DHARMAVATHI,
AND SATYAVATHI DRONAMRAJU

ABSTRACT. We develop a graph signal processing (GSP) framework for analysing atlas-based brain connectomes and for training a geometric deep learning classifier of glioblastoma multiforme (GBM) under an explicit smoothness prior. Structural MRI from two complementary public sources—the healthy IXI cohort and the University of Pennsylvania glioblastoma (UPENN-GBM) collection distributed through The Cancer Imaging Archive—is registered to a common MNI152 reference space and summarised over a shared 48-node Harvard–Oxford cortical atlas graph, so that every scan becomes a signal that can be analysed jointly in the vertex and spectral domains. The graph Fourier transform, Dirichlet energy, spectral centroid, spectral entropy and multiscale spectral graph wavelets are used to characterise disease-related departures from healthy cortical organisation. We prove that the spectral centroid and spectral entropy are invariant to global rescaling of the signal whereas band-limited energies are not, which identifies the former as the descriptors that remain interpretable when intensity scales differ between cohorts. Empirically, the mean GBM T_1 spectrum concentrates its energy near graph frequency $\lambda \approx 0.23$ while the healthy spectrum peaks near $\lambda \approx 0.48$ and retains substantially more mid- and high-frequency content, yielding a lower spectral centroid and lower spectral entropy for GBM; multiscale wavelet energy separates the cohorts most strongly at fine scales. A graph neural network trained under a smoothness-regularised objective—binary cross-entropy prediction loss plus a Dirichlet energy penalty on learned node embeddings—classifies all nine held-out subjects correctly. We show analytically that one gradient step on the smoothness penalty is an explicit Euler step of the graph heat equation, that it is non-expansive precisely when the effective step size does not exceed $1/\lambda_N$, and that it acts as a low-pass filter which suppresses high-frequency noise. We deliberately refrain from presenting the classification result as a validated biomarker: with 28 subjects drawn from two distinct acquisition programmes, cohort and scanner effects cannot be separated from disease effects, and we analyse this confound explicitly. The contribution is a principled, interpretable and reproducible formulation in which the same spectral object—graph Dirichlet energy—serves simultaneously as a descriptive biomarker and as a training-time regulariser.

1. INTRODUCTION

1.1. Motivation. Brain imaging data are intrinsically structured: measurements are not independent across space but are constrained by anatomy, by functional relationships and by long-range organisation [1, 2]. Over the past two decades, network neuroscience has established that the brain is profitably modelled as a graph whose nodes are anatomically defined regions and whose edges encode structural or statistical affinity [3, 4, 5]. Classical descriptors of such networks—small-world organisation [6], efficiency [7] and centrality [4]—characterise the topology itself. When regional quantities are aggregated over a cortical atlas, however, each brain scan additionally defines a *signal supported on the graph nodes*, and the object of interest becomes the interaction between that signal and the topology that carries it. This is precisely the setting of graph signal processing (GSP) [8, 9, 10].

GSP generalises classical harmonic analysis from regular domains (time series, images) to irregular graph domains by adopting the eigenvectors of the graph Laplacian as a Fourier basis [8,

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Corresponding author: Adityakrishna SreeRamachandrarao (asreera110@scientificresearchers.org).

11, 12, 13]. The resulting machinery—the graph Fourier transform, spectral filtering, vertex-frequency analysis [14] and spectral graph wavelets [15, 16]—has proved particularly informative in neuroimaging, where the alignment or misalignment of measured activity with the underlying anatomical network carries functional meaning [17, 18, 19, 20]. Connectome-specific harmonic modes [21] and anatomically adapted graph wavelets [22] illustrate how far the spectral viewpoint can be pushed in this domain.

In parallel, geometric deep learning has extended convolutional architectures to non-Euclidean domains [23, 24], with graph convolutional networks [25], localised spectral filters [26], message-passing formulations [27], attention [28] and inductive sampling [29] now forming a mature toolbox [30, 31]. These models have been applied directly to population and connectome graphs for disease prediction [32, 33, 34]. The present work sits at the intersection of the two threads: we use spectral graph theory both to *describe* disease-related signal structure and to *regularise* a learned classifier, so that a single mathematical object governs the analysis and the inductive bias of the model.

1.2. Clinical Setting. The clinical target is glioblastoma multiforme (GBM), a WHO grade 4 diffuse astrocytic glioma that is the most common and most aggressive primary malignant brain tumour in adults [35, 36]. It accounts for the largest share of malignant primary CNS tumours in population-based registries [37], and prognosis remains poor despite maximal safe resection, radiotherapy and temozolomide chemotherapy [38]. GBM is diffusely infiltrative, molecularly heterogeneous [39] and spatially non-uniform, so a connectome-aware representation that captures *where* and *how* tumour-related signal departs from healthy organisation is well matched to the biology of the disease.

Public imaging resources make such analyses reproducible. The IXI dataset provides nearly 600 healthy-subject multi-sequence MRI volumes [40], while the University of Pennsylvania glioblastoma (UPENN-GBM) cohort distributed through The Cancer Imaging Archive (TCIA) provides multi-parametric MRI with expert tumour segmentations for a large *de novo* GBM population [41, 42, 43].

1.3. Problem Formulation. Let

$$G = (V, E, W)$$

denote an undirected weighted graph with node set $V = \{1, \dots, N\}$, edge set E and symmetric weighted adjacency matrix $W \in \mathbb{R}^{N \times N}$ with $W_{ij} \geq 0$. For each subject s , an imaging-derived regional measurement is represented as a graph signal

$$x^{(s)} \in \mathbb{R}^N,$$

whose i -th entry is the mean signal value in region i . A disease classifier maps the graph signal, and optionally the graph structure, to a label

$$y^{(s)} \in \{0, 1\},$$

with $y^{(s)} = 1$ denoting GBM and $y^{(s)} = 0$ denoting a healthy subject.

The central modelling principle, inherited from spectral graph theory and from semi-supervised learning on graphs [44, 45, 46], is that biologically meaningful graph signals are typically *piecewise smooth*: neighbouring regions take related values except where pathology disrupts them. This motivates regularising the learned predictive representation by graph Dirichlet energy, giving an objective that balances prediction fidelity against structural regularity,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{smooth}}, \quad (1.1)$$

where $\mathcal{L}_{\text{pred}}$ is task-specific, $\mathcal{L}_{\text{smooth}}$ is a Dirichlet energy penalty and $\lambda > 0$ is the regularisation weight.

1.4. Contributions. This paper makes the following contributions.

- (i) We construct a shared 48-node Harvard–Oxford atlas graph in MNI152 space that gives every node an identical anatomical interpretation across a healthy cohort and a disease cohort, and therefore a single spectral basis for all subsequent analysis (Sections 3.2–3.3).

- (ii) We define T_1 -intensity and tumour-burden graph signals and characterise disease-related departures using the graph Fourier transform, Dirichlet energy, spectral centroid, spectral entropy and multiscale spectral graph wavelets (Sections 3.4–3.6).
- (iii) We prove that the spectral centroid and spectral entropy are invariant under global rescaling of a graph signal, whereas band-limited energies are not (Proposition 3.3). This is of direct practical consequence when cohorts are acquired on different scanners, and it identifies which descriptors may legitimately be compared across cohorts without intensity harmonisation.
- (iv) We show that a gradient step on the Dirichlet penalty is an explicit Euler step of the graph heat equation, that it is a low-pass spectral filter, and that it is non-expansive exactly when the effective step size does not exceed $1/\lambda_N$ (Propositions 3.6 and 3.7). This converts a heuristic regulariser into an analysable operator and yields a concrete admissible range for λ .
- (v) We report spectral, wavelet and classification results on the combined cohort, and we analyse—rather than understate—the cohort, acquisition and sample-size confounds that limit what the classification result can support (Section 5.4).

1.5. Outline. Section 2 situates the work in the literature. Section 3 describes the imaging cohorts, preprocessing, graph construction, spectral and wavelet operators, the diffusion interpretation and the geometric deep learning model. Section 4 reports descriptive spectral biomarkers and classifier performance. Section 5 interprets the findings and states limitations, and Section 6 concludes. Notation is summarised in Table 3.

2. RELATED WORK

Graph signal processing. The spectral theory of the graph Laplacian is classical [11, 12, 13, 47]. Its reformulation as a signal processing framework is comparatively recent [8, 10, 9], and now includes multiscale wavelets [15, 16], vertex-frequency localisation [14], stationarity theory [48] and methods for inferring the graph itself from smooth observations [49, 50, 51]. We take the graph as given by an anatomical atlas and concentrate on the signal and the learning problem defined over it.

Spectral analysis of brain signals. Treating cortical measurements as graph signals has clarified how function couples to structure [17, 18, 19]. Atasoy et al. [21] interpret Laplacian eigenmodes as connectome-specific harmonic waves; Preti and Van De Ville [20] use the split between low- and high-frequency graph content to quantify regional structure–function decoupling; Behjat et al. [22] design anatomically adapted graph wavelets for group-level activation mapping. Our spectral descriptors follow this tradition, with the addition of an explicit invariance analysis (Proposition 3.3) that we have not seen stated in this setting.

Geometric deep learning for neuroimaging. Graph convolutional networks [25, 26] have been applied to population graphs for Autism Spectrum Disorder and Alzheimer’s disease [32], to metric learning over connectivity networks [33] and to interpretable region-level fMRI models [34]. A recurring theme in the broader GNN literature is that repeated neighbourhood averaging behaves as a low-pass filter [52, 53] and, taken far enough, collapses representations—the oversmoothing phenomenon [54, 55, 56]. Our Dirichlet penalty is an *explicit* and tunable form of the same smoothing pressure; Section 3.8.6 discusses why making it explicit is preferable to leaving it implicit in the depth of the network.

Glioblastoma imaging. Expert-annotated public GBM collections and the BraTS benchmark have standardised tumour sub-region definitions [57, 58, 59], and the UPENN-GBM release adds clinical, genomic and radiomic annotation [41]. We use these segmentations only to define node-wise tumour burden; we do not attempt segmentation itself.

3. MATERIALS AND METHODS

3.1. Imaging Cohorts. Two complementary, publicly available structural MRI resources define the healthy and disease cohorts. Using established public references on both sides ensures that the analysis is reproducible and that node semantics are shared across groups.

Healthy cohort (IXI). Healthy-brain reference signals are drawn from the IXI dataset (“Information eXtraction from Images”), a collection of nearly 600 MRI examinations of normal, healthy subjects acquired at three London hospitals, comprising T_1 -, T_2 - and proton-density-weighted, MRA and diffusion-weighted images [40]. Because IXI subjects are free of focal pathology, their atlas-wise intensity signals provide a baseline model of typical, smoothly varying cortical organisation against which disease-related departures are measured.

Disease cohort (UPENN-GBM via TCIA). Glioblastoma signals are drawn from the University of Pennsylvania glioblastoma cohort [41], distributed by The Cancer Imaging Archive [43, 42]. UPENN-GBM provides multi-parametric MRI (pre- and post-contrast T_1 -weighted, T_2 -weighted and T_2 -FLAIR) for *de novo* GBM patients together with expert-reviewed segmentation labels demarcating the enhancing tumour, the peritumoural oedema and the non-enhancing core. These labels follow the annotation conventions established by the BraTS benchmark and the TCGA glioma collections [57, 58] and yield the voxel-level tumour mask $M^{(s)}$ used below.

Analysed sample. The analyses reported in Section 4 use 28 subjects in total, partitioned into 19 training and 9 held-out test subjects; the held-out split comprises 6 healthy and 3 GBM subjects. This is a small sample, and Section 5.4 examines in detail what can and cannot be inferred from it.

Remark 3.1. The two cohorts originate from entirely separate acquisition programmes, with different scanners, field strengths and protocols. Any between-cohort contrast therefore confounds disease with site. We retain this design because it is the only configuration supported by the available public data, but we treat it as a first-order limitation throughout rather than as a nuisance detail; see Sections 4.4 and 5.4.

3.2. Preprocessing and Spatial Registration. To compare subjects on a common node set, all volumes are mapped into a standard stereotaxic space. The nonlinear MNI152 template [60] serves as the common reference frame, and the Harvard–Oxford cortical atlas—derived from expert manual segmentations at the Center for Morphometric Analysis [61], and conceptually related to automated gyral parcellations of the cortex [62] and to other widely used parcellations [63, 64]—supplies the ROI labels that become graph nodes. Registration of subject images and tumour masks to MNI space follows standard neuroimaging practice, combining intensity inhomogeneity correction [65] with affine and nonlinear alignment as implemented in FSL [66, 67] and ANTs [68].

The atlas-based construction pipeline is illustrated in Figures 1–3. The MNI152 template provides the common reference space, the Harvard–Oxford atlas provides the ROI labels, and subject-level MRI volumes and tumour masks are mapped into this space so that node-wise quantities can be computed consistently across the cohort. After registration, each atlas node receives a region-level summary statistic, such as the mean T_1 intensity or the fraction of voxels covered by tumour.

3.3. Construction of the Connectome Graph.

3.3.1. Network definition. A common atlas provides a shared node set across subjects. The Harvard–Oxford cortical atlas defines $N = 48$ cortical ROIs and hence a fixed graph topology for all subjects, so the node set is $V = \{1, \dots, 48\}$. An undirected edge $(i, j) \in E$ is included when regions i and j are regarded as anatomically or statistically connected in the shared reference graph. The graph is represented by its weighted adjacency matrix W , its degree matrix D and its combinatorial Laplacian L :

$$W_{ij} = W_{ji} \geq 0, \quad D_{ii} = \sum_{j=1}^N W_{ij}, \quad L = D - W. \quad (3.1)$$

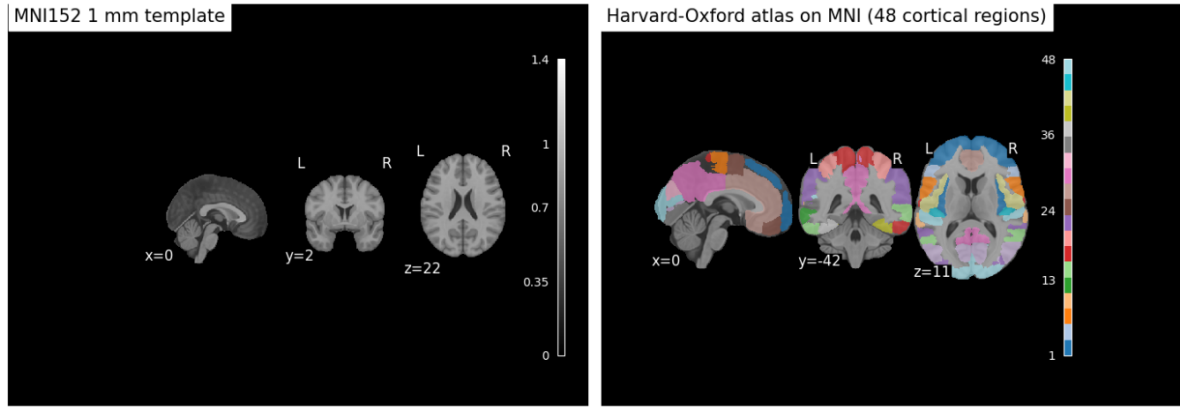


Figure 1. Reference space and atlas definition. Left: MNI152 1 mm template [60]. Right: the Harvard–Oxford atlas [61] projected onto MNI space with 48 cortical regions. This atlas defines the graph nodes used throughout the analysis.

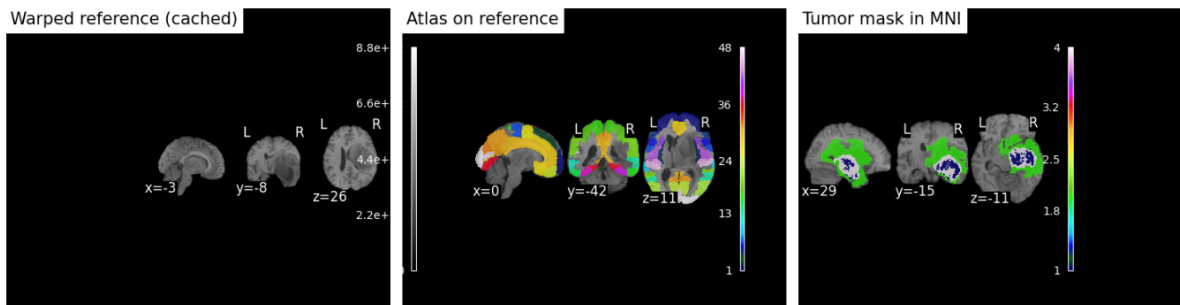


Figure 2. Registration and atlas alignment workflow. The reference image, the atlas labels and the tumour mask are shown in common space, enabling atlas-wise aggregation of tumour burden and of image-derived regional signals.

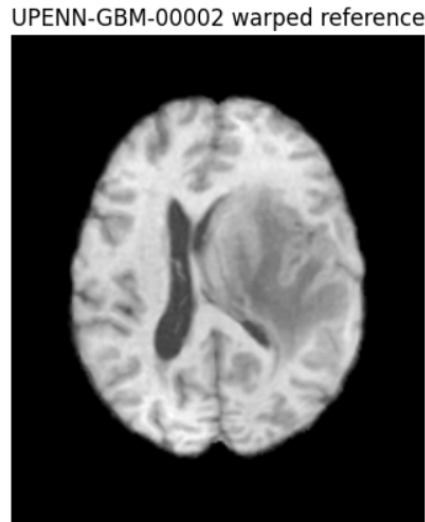


Figure 3. Example subject in aligned space. A representative GBM case is shown after warping into the common reference frame used for atlas aggregation.

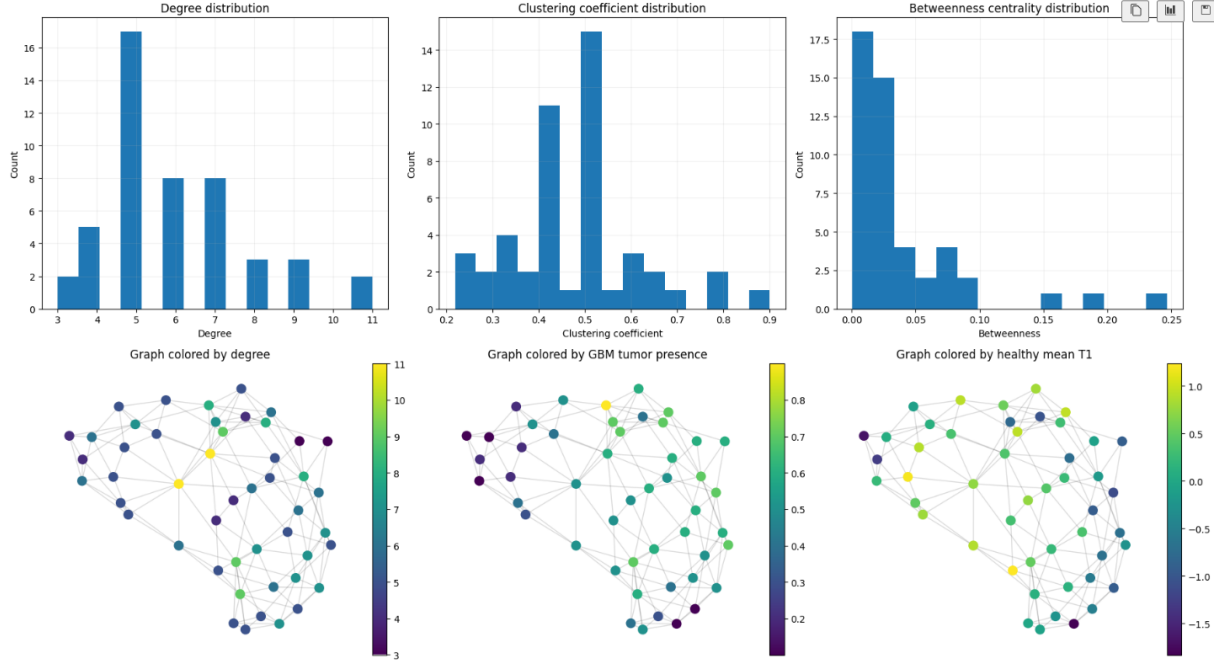


Figure 4. Descriptive graph statistics and graph visualisations for the shared 48-node connectome. Top row: distributions of degree, clustering coefficient and betweenness centrality. Bottom row: the same graph coloured by degree, by average GBM tumour presence and by average healthy mean T_1 signal.

Because node degrees vary, the symmetric normalised Laplacian [11] is also useful:

$$L_{\text{sym}} = I - D^{-1/2}WD^{-1/2}. \quad (3.2)$$

The normalised form permits comparison across regions with different connectivity levels, has spectrum contained in $[0, 2]$, and is the operator underlying standard graph convolutions [25, 26].

3.3.2. Descriptive topology of the shared connectome. Standard network-neuroscience descriptors [4, 5] summarise the topology of the shared graph. Figure 4 reports the degree distribution, the clustering coefficient distribution, the betweenness centrality distribution, and graph renderings coloured by several node-level attributes. If d_i denotes the degree of node i , the average degree is

$$\bar{d} = \frac{1}{N} \sum_{i=1}^N d_i, \quad (3.3)$$

the local clustering coefficient of node i is

$$C_i = \frac{2T_i}{d_i(d_i - 1)}, \quad (3.4)$$

where T_i is the number of triangles through node i , and betweenness centrality quantifies how frequently a node lies on shortest paths,

$$B_i = \sum_{\substack{s \neq i \neq t \\ s < t}} \frac{\sigma_{st}(i)}{\sigma_{st}}, \quad (3.5)$$

with σ_{st} the number of shortest s - t paths and $\sigma_{st}(i)$ the number of those passing through i [4]. These properties are not merely descriptive: a penalty of the form $f^\top Lf$ weights each edge separately, so nodes with many strong incident edges contribute disproportionately to the total regularisation. The overall strength of the penalty is in turn governed by the Laplacian spectrum, whose smallest non-trivial eigenvalue is the algebraic connectivity of the graph [47].

3.4. Brain Measurements as Graph Signals.

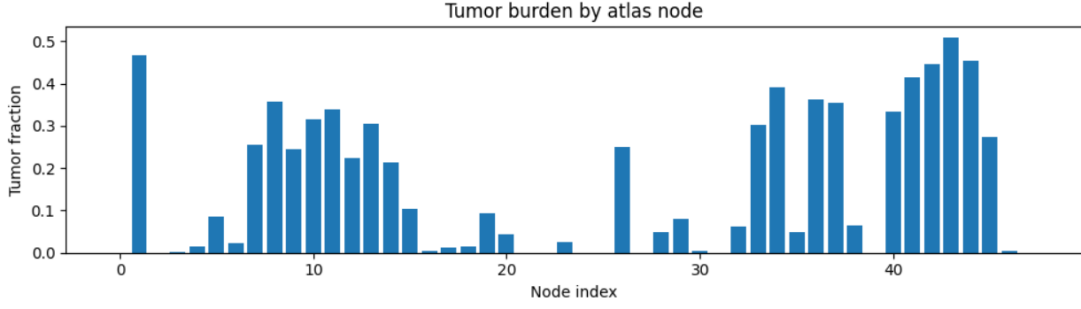


Figure 5. Tumour burden by atlas node for a representative GBM subject. The node-wise tumour fraction defines a graph signal that can be analysed both spectrally and spatially.

3.4.1. *Regional signal definition.* Given a subject-level image $I^{(s)}(\mathbf{r})$ in common space and an atlas region Ω_i , the node-wise regional summary signal is

$$x_i^{(s)} = \frac{1}{\mu(\Omega_i)} \int_{\Omega_i} I^{(s)}(\mathbf{r}) d\mathbf{r}, \quad (3.6)$$

where $\mu(\Omega_i)$ is the volume of region i . Stacking across nodes yields the graph signal $x^{(s)} \in \mathbb{R}^N$. Two signals are of particular interest:

- (a) the mean T_1 signal $x_{t1}^{(s)}$, computed from the IXI (healthy) and UPENN-GBM (disease) T_1 -weighted volumes; and
- (b) the tumour burden signal $x_{\text{tumor}}^{(s)}$, derived from the UPENN-GBM expert segmentations [41], with

$$x_{\text{tumor},i}^{(s)} = \frac{1}{|\Omega_i|} \sum_{\mathbf{r} \in \Omega_i} M^{(s)}(\mathbf{r}), \quad (3.7)$$

where $M^{(s)}(\mathbf{r}) \in \{0, 1\}$ is the tumour mask.

These two signals are of different physical types: x_{t1} is an intensity in arbitrary scanner units, whereas x_{tumor} is a dimensionless volume fraction in $[0, 1]$. Quantitative comparisons are therefore made *within* signal type—GBM T_1 against healthy T_1 —and the tumour signal is used as a descriptive reference. Proposition 3.3 makes precise which descriptors are insensitive to this distinction.

Figure 5 shows the tumour burden by atlas node for a representative case, revealing a spatially non-uniform disease distribution over the connectome that is consistent with the diffusely infiltrative nature of GBM [35, 36].

3.4.2. *Cohort mean signals and disease contrast.* Let $\mathcal{S}_{\text{GBM}}$ and $\mathcal{S}_{\text{H}}$ denote the GBM and healthy cohorts. Their mean signals and the induced disease contrast are

$$\bar{x}_{\text{GBM}} = \frac{1}{|\mathcal{S}_{\text{GBM}}|} \sum_{s \in \mathcal{S}_{\text{GBM}}} x^{(s)}, \quad \bar{x}_{\text{H}} = \frac{1}{|\mathcal{S}_{\text{H}}|} \sum_{s \in \mathcal{S}_{\text{H}}} x^{(s)}, \quad \Delta x = \bar{x}_{\text{GBM}} - \bar{x}_{\text{H}}. \quad (3.8)$$

3.5. Spectral Graph Analysis.

3.5.1. *Graph Fourier transform.* The Laplacian L is real symmetric and positive semidefinite, so by the spectral theorem [69] it admits the eigendecomposition [11, 8]

$$L = U\Lambda U^\top, \quad U = [u_1, \dots, u_N], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_N), \quad (3.9)$$

with $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$ and U orthogonal. The second-smallest eigenvalue λ_2 is the algebraic connectivity of the graph [47], and the eigenvectors u_k generalise the classical Fourier harmonics to the irregular graph domain [45, 12]. The graph Fourier transform (GFT) and its inverse are [8, 10]

$$\hat{x} = U^\top x, \quad x = U\hat{x} = \sum_{k=1}^N \hat{x}_k u_k. \quad (3.10)$$

Low graph frequencies correspond to eigenvectors with small eigenvalues, which vary slowly across strongly connected regions; high graph frequencies oscillate sharply between neighbouring nodes.

3.5.2. Dirichlet energy. The central quantity of the paper is the graph Dirichlet energy [8, 44],

$$\mathcal{E}(x) = x^\top Lx = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N W_{ij} (x_i - x_j)^2. \quad (3.11)$$

Proposition 3.2 (Vertex and spectral forms of Dirichlet energy). *For every $x \in \mathbb{R}^N$,*

$$\mathcal{E}(x) = \frac{1}{2} \sum_{i,j} W_{ij} (x_i - x_j)^2 = \sum_{k=1}^N \lambda_k |\hat{x}_k|^2. \quad (3.12)$$

In particular $\mathcal{E}(x) \geq 0$, with $\mathcal{E}(x) = 0$ if and only if x is constant on each connected component of G .

Proof. Expanding the quadratic form and using $D_{ii} = \sum_j W_{ij}$,

$$\frac{1}{2} \sum_{i,j} W_{ij} (x_i - x_j)^2 = \sum_i D_{ii} x_i^2 - \sum_{i,j} W_{ij} x_i x_j = x^\top (D - W)x = x^\top Lx.$$

Substituting $L = U\Lambda U^\top$ and $\hat{x} = U^\top x$ gives $x^\top Lx = \hat{x}^\top \Lambda \hat{x} = \sum_k \lambda_k |\hat{x}_k|^2$. Since $\lambda_k \geq 0$ the form is non-negative, and it vanishes exactly when $W_{ij}(x_i - x_j)^2 = 0$ for all i, j , that is, when x is constant on each connected component. $\square$

Equation (3.12) is the formal statement that the two readings of smoothness coincide: in the vertex domain Dirichlet energy penalises edge-wise disagreement, and in the spectral domain it weights signal energy by graph frequency. A signal is smooth precisely when its spectral mass sits at small eigenvalues [45].

3.5.3. Spectral biomarkers. Several scalar descriptors summarise a graph signal compactly, in the spirit of GSP-based analysis of brain imaging [17, 18].

Spectral centroid. The energy-weighted mean graph frequency,

$$C_{\text{spec}}(x) = \frac{\sum_{k=1}^N \lambda_k |\hat{x}_k|^2}{\sum_{k=1}^N |\hat{x}_k|^2}, \quad (3.13)$$

locates the signal in graph frequency. Larger values indicate a shift towards less smooth components.

Spectral entropy. With normalised spectral weights $p_k = |\hat{x}_k|^2 / \sum_\ell |\hat{x}_\ell|^2$, the spectral entropy is

$$H_{\text{spec}}(x) = - \sum_{k=1}^N p_k \log p_k. \quad (3.14)$$

Larger values indicate energy distributed more evenly across graph harmonics; H_{spec} attains its maximum $\log N$ for a flat spectrum and vanishes when all energy sits in a single mode.

Band-limited energies. Let $\mathcal{K}_{\text{low}}$ and $\mathcal{K}_{\text{high}}$ index low and high graph frequencies. Then

$$E_{\text{low}}(x) = \sum_{k \in \mathcal{K}_{\text{low}}} |\hat{x}_k|^2, \quad E_{\text{high}}(x) = \sum_{k \in \mathcal{K}_{\text{high}}} |\hat{x}_k|^2. \quad (3.15)$$

Larger E_{high} indicates sharper spatial variation and reduced smoothness. In the results below, $\mathcal{K}_{\text{high}}$ is the upper 20% of the spectrum and the energies are normalised by $\|x\|_2^2$, so that $E_{\text{low}} + E_{\text{high}} = 1$.

The following elementary but practically important observation distinguishes the two families of descriptors.

Table 1. Spectral biomarkers and their interpretation in the connectome graph setting. The last column records behaviour under the global rescaling $x \mapsto cx$ (Proposition 3.3).

Metric	Definition	Interpretation	Rescaling
Low-frequency energy	$E_{\text{low}} = \sum_{k \in \mathcal{K}_{\text{low}}} \hat{x}_k ^2$	Signal power carried by smooth graph harmonics; larger values imply stronger global smoothness.	$c^2 E_{\text{low}}$
High-frequency energy	$E_{\text{high}} = \sum_{k \in \mathcal{K}_{\text{high}}} \hat{x}_k ^2$	Signal power carried by rapidly varying harmonics; larger values imply sharper regional contrast.	$c^2 E_{\text{high}}$
Spectral centroid	$C_{\text{spec}} = \frac{\sum_k \lambda_k \hat{x}_k ^2}{\sum_k \hat{x}_k ^2}$	Energy-weighted mean graph frequency; higher values indicate a spectrum shifted towards less smooth components.	invariant
Spectral entropy	$H_{\text{spec}} = -\sum_k p_k \log p_k$	Dispersion of the spectrum across modes; higher values indicate energy distributed more evenly.	invariant

Proposition 3.3 (Scale invariance of centroid and entropy). *Let $x \in \mathbb{R}^N \setminus \{0\}$ and let $c \neq 0$. Then*

$$C_{\text{spec}}(cx) = C_{\text{spec}}(x), \quad H_{\text{spec}}(cx) = H_{\text{spec}}(x), \quad (3.16)$$

whereas the unnormalised band energies obey $E_{\text{low}}(cx) = c^2 E_{\text{low}}(x)$ and $E_{\text{high}}(cx) = c^2 E_{\text{high}}(x)$.

Proof. The GFT is linear, so $\widehat{cx} = c\hat{x}$ and $|\widehat{cx}_k|^2 = c^2 |\hat{x}_k|^2$. The factor c^2 appears in both the numerator and the denominator of (3.13) and cancels, so C_{spec} is unchanged. Likewise the normalised weights satisfy $p_k(cx) = c^2 |\hat{x}_k|^2 / \sum_\ell c^2 |\hat{x}_\ell|^2 = p_k(x)$, so H_{spec} is unchanged. The band energies are sums of $|\hat{x}_k|^2$ without normalisation and therefore scale by c^2 . $\square$

Remark 3.4. MRI intensities are expressed in arbitrary units that depend on scanner, sequence and reconstruction, which is why intensity standardisation is a routine preprocessing step [70, 65]. Proposition 3.3 shows that C_{spec} and H_{spec} are immune to a global gain difference between cohorts, whereas raw band energies are not. The centroid and the entropy are therefore the descriptors that may be compared across cohorts with the fewest assumptions; the normalised energies used in Section 4 inherit the same invariance by construction, since dividing by $\|x\|_2^2$ removes the common factor c^2 . Invariance to a global gain does not, however, confer invariance to the non-affine contrast differences that distinguish scanners, and we return to this point in Section 5.4.

The interpretation of each descriptor is summarised in Table 1.

3.6. Graph Wavelet Analysis. Scalar spectral descriptors summarise the global distribution of graph frequency content but discard localisation in the vertex domain. Spectral graph wavelets restore it by defining multiscale filters on the Laplacian spectrum [15, 16, 14].

Let $g(\cdot)$ be a spectral kernel. The graph wavelet operator at scale $\tau > 0$ is [15]

$$T_g^{(\tau)} = g(\tau L) = U g(\tau \Lambda) U^\top, \quad g(\tau \Lambda) = \text{diag}(g(\tau \lambda_1), \dots, g(\tau \lambda_N)), \quad (3.17)$$

and the wavelet coefficients of a signal x at scale τ are $w^{(\tau)} = T_g^{(\tau)} x$. For heat-kernel diffusion wavelets one takes $g(\tau \lambda) = e^{-\tau \lambda}$, which ties the construction directly to diffusion geometry [71]: small τ emphasises fine detail and large τ emphasises coarse structure. By orthogonality of U , the scale-dependent wavelet energy is

$$E_{\text{wav}}(\tau; x) = \|w^{(\tau)}\|_2^2 = \sum_{k=1}^N |g(\tau \lambda_k)|^2 |\hat{x}_k|^2. \quad (3.18)$$

Equation (3.18) shows that wavelet energy is a reweighting of the same spectral mass that defines $\mathcal{E}$, C_{spec} and H_{spec} , with the kernel $g(\tau \cdot)$ selecting the band under examination. This is the bridge

between the descriptive multiscale analysis of Section 4.3 and the graph-regularised learning problem of Section 3.8.

3.7. Diffusion Dynamics and Dirichlet Smoothness.

3.7.1. Heat diffusion on the graph. The standard physical reading of graph smoothness is diffusion [71]. Let $f(t) \in \mathbb{R}^N$ satisfy the graph heat equation

$$\frac{df(t)}{dt} = -Lf(t), \quad f(0) = f_0, \quad (3.19)$$

whose solution is

$$f(t) = e^{-tL} f_0 = U e^{-t\Lambda} U^\top f_0. \quad (3.20)$$

Each spectral component decays as $e^{-t\lambda_k}$, so high-frequency content vanishes faster than low-frequency content.

Lemma 3.5 (Dirichlet energy as dissipation rate). *If f solves (3.19), then*

$$\frac{d}{dt} \|f(t)\|_2^2 = -2\mathcal{E}(f(t)) \leq 0. \quad (3.21)$$

Proof. Differentiating and using symmetry of L , $\frac{d}{dt} \|f\|_2^2 = 2f^\top \dot{f} = -2f^\top Lf = -2\mathcal{E}(f)$, which is non-positive by Proposition 3.2. $\square$

Lemma 3.5 gives Dirichlet energy an operational meaning: it is the instantaneous rate at which diffusion dissipates signal energy. Signals with large Dirichlet energy contain sharp edge-wise discrepancies and smooth rapidly; signals with small Dirichlet energy are already smooth and evolve slowly.

3.7.2. Interpretation for brain signals. The smoothness assumption does *not* assert that disease effects are spatially uniform. It asserts that the *learned predictive representation* should avoid node-to-node oscillation unsupported by the graph, in line with manifold- and graph-regularised learning [45, 44]. The goal is not to maximise smoothness but to minimise $\mathcal{L}_{\text{pred}} + \lambda\mathcal{L}_{\text{smooth}}$, so that the variation which survives is the variation the task requires.

3.8. Geometric Deep Learning Model.

3.8.1. Model overview. Given the graph $G = (V, E, W)$ and a subject signal $x^{(s)}$, the task is to predict cohort membership. Following the geometric deep learning paradigm [23, 24, 30], a graph neural network maps node features to a graph-level representation and then to a class probability,

$$\hat{y}^{(s)} = \sigma(\phi_{\text{readout}}(H^{(L,s)})), \quad (3.22)$$

where $H^{(L,s)} \in \mathbb{R}^{N \times F_L}$ is the node embedding matrix after L layers, ϕ_{readout} is a permutation-invariant graph-level pooling function and σ is the logistic sigmoid.

3.8.2. Message passing and graph convolution. A general message passing layer is [27]

$$h_i^{(\ell+1)} = \psi^{(\ell)} \left(h_i^{(\ell)}, \bigoplus_{j \in \mathcal{N}(i)} \phi^{(\ell)}(h_i^{(\ell)}, h_j^{(\ell)}, e_{ij}) \right), \quad (3.23)$$

where $\oplus$ is a permutation-invariant aggregation. Specialising to the normalised-adjacency graph convolution [25],

$$H^{(\ell+1)} = \rho \left(\tilde{D}^{-1/2} \tilde{W} \tilde{D}^{-1/2} H^{(\ell)} \Theta^{(\ell)} \right), \quad (3.24)$$

where $\tilde{W} = W + I$, $\tilde{D}$ is the corresponding degree matrix, $\Theta^{(\ell)}$ are learnable weights and ρ is a pointwise nonlinearity. Equation (3.24) is a first-order localised spectral filter [26], and it makes explicit how the graph topology constrains feature mixing between neighbouring regions.

Table 2. Components of the training objective and their roles.

Component	Definition	Purpose
Prediction loss	$\mathcal{L}_{\text{pred}} = -\frac{1}{M} \sum_s [y^{(s)} \log \hat{p}^{(s)} + (1 - y^{(s)}) \log(1 - \hat{p}^{(s)})]$	Drives accurate subject-level cohort classification.
Smoothness penalty	$\mathcal{L}_{\text{smooth}} = \frac{1}{M} \sum_s \text{Tr}((H^{(L,s)})^\top L H^{(L,s)})$	Penalises edge-wise discrepancies between learned node embeddings.
Total objective	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{smooth}}$	Trades classification fidelity against graph-structured regularity.

3.8.3. *Graph-level readout.* The graph-level embedding is obtained by global mean pooling,

$$z^{(s)} = \frac{1}{N} \sum_{i=1}^N h_i^{(L,s)}, \quad (3.25)$$

and the predicted probability is

$$\hat{p}^{(s)} = \sigma(w^\top z^{(s)} + b) = \frac{1}{1 + e^{-\eta^{(s)}}}, \quad \eta^{(s)} = w^\top z^{(s)} + b. \quad (3.26)$$

3.8.4. *Training objective.* The prediction loss is binary cross-entropy over the M training subjects,

$$\mathcal{L}_{\text{pred}} = -\frac{1}{M} \sum_{s=1}^M [y^{(s)} \log \hat{p}^{(s)} + (1 - y^{(s)}) \log(1 - \hat{p}^{(s)})]. \quad (3.27)$$

The smoothness penalty is the matrix-valued generalisation of Dirichlet energy over learned embeddings, extending scalar Laplacian regularisation [44, 45] to vector-valued node features:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{M} \sum_{s=1}^M \text{Tr}((H^{(L,s)})^\top L H^{(L,s)}) = \frac{1}{2M} \sum_{s=1}^M \sum_{i,j} W_{ij} \|h_i^{(L,s)} - h_j^{(L,s)}\|_2^2. \quad (3.28)$$

Neighbouring nodes with strong graph affinity are thereby encouraged to acquire similar embeddings unless task gradients force them apart. The components of the objective are collected in Table 2.

3.8.5. *The smoothness gradient is a diffusion step.* The following two results make the regulariser analysable rather than heuristic.

Proposition 3.6 (Gradient flow of the penalty is graph diffusion). *Let $H \in \mathbb{R}^{N \times F}$ and $S(H) = \text{Tr}(H^\top L H)$. Then $\nabla_H S(H) = 2LH$, and a gradient descent step on λS with learning rate $\eta > 0$ is*

$$H \mapsto H - 2\eta\lambda LH = (I - 2\eta\lambda L)H, \quad (3.29)$$

which is exactly the explicit Euler discretisation of the graph heat equation (3.19) with time step $\Delta t = 2\eta\lambda$.

Proof. Writing $S(H) = \sum_{f=1}^F H_{:,f}^\top L H_{:,f}$ and differentiating each quadratic form with L symmetric gives $\partial S / \partial H_{:,f} = 2LH_{:,f}$, hence $\nabla_H S = 2LH$. The update $H \mapsto H - \eta \nabla_H (\lambda S)$ is then (3.29). Comparing with the forward difference $H(t + \Delta t) \approx H(t) - \Delta t LH(t)$ applied to (3.19) identifies $\Delta t = 2\eta\lambda$. $\square$

Proposition 3.7 (Low-pass action and stability). *The update (3.29) acts in the graph Fourier domain as the pointwise multiplier*

$$\hat{h}_k \mapsto (1 - 2\eta\lambda \lambda_k) \hat{h}_k, \quad k = 1, \dots, N. \quad (3.30)$$

Its operator norm satisfies $\|I - 2\eta\lambda L\|_2 \leq 1$ if and only if

$$0 < \eta\lambda \leq \frac{1}{\lambda_N}, \quad (3.31)$$

and under (3.31) the multiplier is non-increasing in λ_k , so the step attenuates high graph frequencies at least as strongly as low ones and leaves the constant mode $\lambda_1 = 0$ untouched.

Proof. Applying $U^\top$ to (3.29) and using $L = U\Lambda U^\top$ gives $U^\top H \mapsto (I - 2\eta\lambda\Lambda)U^\top H$, which is the stated pointwise multiplier. The matrix $I - 2\eta\lambda L$ is symmetric with eigenvalues $1 - 2\eta\lambda\lambda_k$, so its spectral norm is $\max_k |1 - 2\eta\lambda\lambda_k|$. This is at most 1 exactly when $0 \leq 2\eta\lambda\lambda_k \leq 2$ for every k , that is, when $\eta\lambda\lambda_N \leq 1$. Under this condition $1 - 2\eta\lambda\lambda_k$ is non-negative and decreasing in λ_k , and equals 1 at $\lambda_1 = 0$. $\square$

Remark 3.8. Proposition 3.7 supplies a concrete admissible range for the regularisation weight: given a learning rate η , choosing $\lambda \leq 1/(\eta\lambda_N)$ guarantees that the smoothing component of the update cannot amplify any spectral mode. Exceeding this bound makes the smoothing step expansive on the highest-frequency modes, so a penalty intended to suppress oscillation would instead inject it. For the normalised Laplacian L_{sym} one has $\lambda_N \leq 2$ [11], which yields the convenient sufficient condition $\eta\lambda \leq 1/2$.

3.8.6. Relation to oversmoothing. Proposition 3.7 also clarifies the relationship between this penalty and the oversmoothing phenomenon. Repeated neighbourhood averaging in deep GNNs is itself a low-pass operation [52, 53], and iterating it drives node representations towards a low-dimensional invariant subspace, degrading discriminability [54, 55]. Mitigations such as edge dropping [56] address the symptom. The formulation adopted here differs in that the smoothing pressure is an explicit additive term with a single interpretable coefficient λ and a known stability bound (3.31), rather than an implicit consequence of depth. The network can therefore be kept shallow, and the amount of smoothing can be selected—and reported—independently of the architecture.

3.9. Implementation and Reproducibility. Graph construction and topological descriptors were computed with NetworkX [72]; atlas handling and neuroimaging-specific operations used the nilearn/scikit-learn stack [73]; numerical linear algebra, including the Laplacian eigendecomposition in (3.9) and the matrix exponential in (3.20), used NumPy [74] and SciPy [75]. The graph neural network was implemented in PyTorch [76] with PyTorch Geometric [77] and trained with Adam [78] for 150 epochs on the 19-subject training split, with the objective (1.1) evaluated on the full graph for every subject. Because $N = 48$, the Laplacian spectrum is computed exactly and no polynomial approximation of the spectral filters is required [26].

4. RESULTS

4.1. Spatial Distribution of Disease Effects. Figure 6 shows the cohort mean graph signals and their difference Δx over the shared connectome, and Figure 7 shows the average tumour presence projected onto the same graph. The disease effect is not spatially uniform: it concentrates over subsets of regions rather than spreading evenly across the cortex. This structured, localised character is what motivates multiscale analysis on the graph [15, 22] rather than a single global summary.

4.2. Spectral Biomarkers. Applying the GFT to the cohort mean signals reveals a systematic redistribution of spectral energy between groups. Figure 8 compares the mean T_1 spectra. The GBM spectrum is dominated by a single sharp peak near $\lambda \approx 0.23$ reaching a mean spectral power of approximately 16, whereas the healthy spectrum peaks later, near $\lambda \approx 0.48$, at approximately 9.6, and retains appreciable secondary structure at $\lambda \approx 0.75$, 0.91 and 1.05. The GBM signal is thus both more concentrated and positioned lower in graph frequency than the healthy signal: energy that is distributed across several mid-frequency harmonics in healthy brains is consolidated into one low-frequency mode in the disease cohort. This is consistent with a diffusely infiltrative lesion that imposes a large, spatially coherent perturbation on the cortical signal [35, 36].

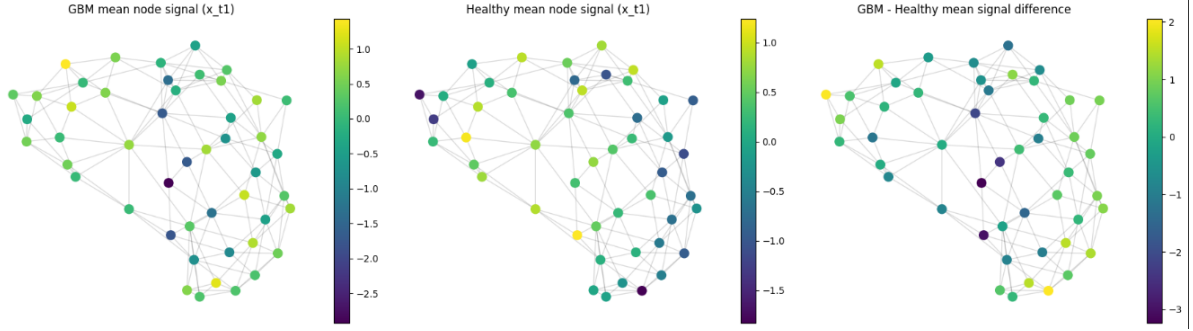


Figure 6. Cohort mean graph signals and their difference. Left: mean GBM node signal. Middle: mean healthy node signal. Right: the GBM minus healthy contrast Δx . Disease-related deviations are spatially structured over the graph rather than randomly distributed.

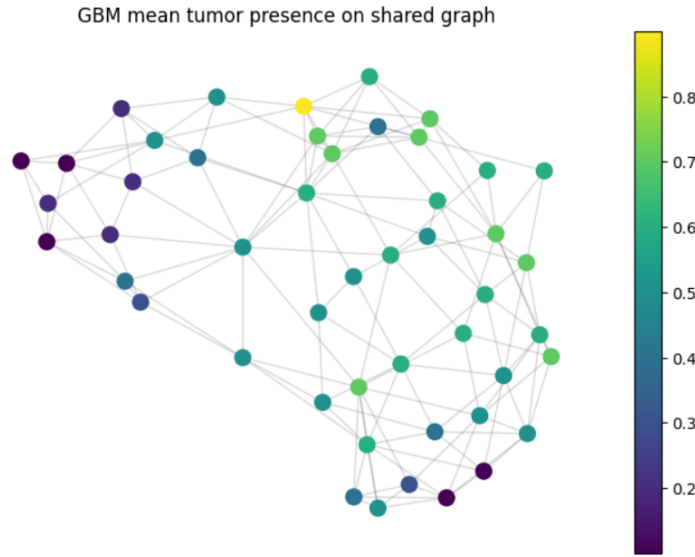


Figure 7. Average GBM tumour presence projected onto the shared graph. This cohort-level map indicates which atlas nodes are most frequently involved by tumour across GBM subjects.

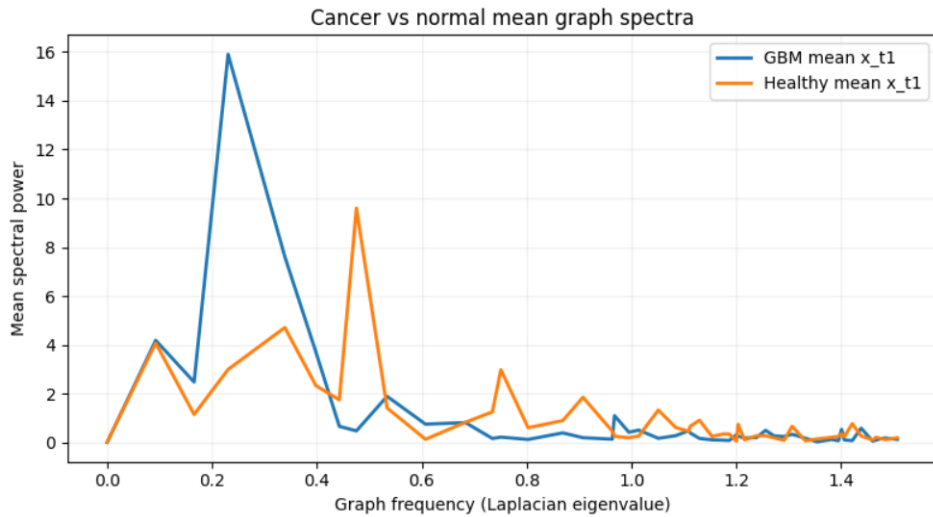


Figure 8. Mean graph spectra of the T_1 signal for the GBM and healthy cohorts, plotted as mean spectral power against graph frequency (Laplacian eigenvalue). The GBM spectrum concentrates near $\lambda \approx 0.23$ while the healthy spectrum peaks near $\lambda \approx 0.48$ and retains more mid-frequency structure.

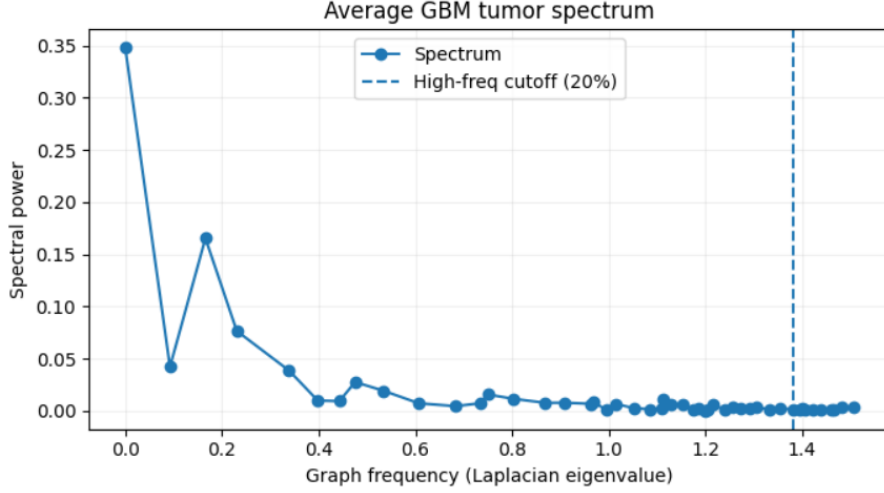


Figure 9. Average GBM tumour spectrum. The dashed vertical line marks the high-frequency cutoff, placed at the upper 20% of graph frequencies, that defines the index set $\mathcal{K}_{\text{high}}$ used for the band-limited energies.

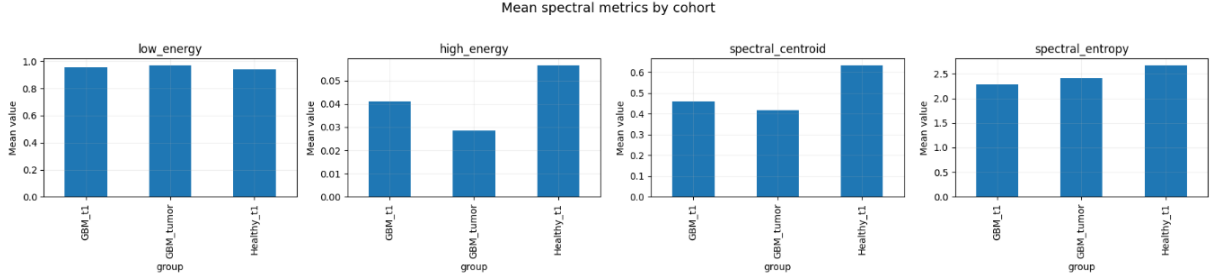


Figure 10. Mean spectral metrics by group: normalised low-frequency energy, normalised high-frequency energy, spectral centroid and spectral entropy, for the GBM T_1 , GBM tumour and healthy T_1 signals. Low-frequency energy is saturated near 0.95 in all groups; the discriminative contrast resides in the remaining three descriptors.

Figure 10 summarises the four scalar descriptors for three signal groups: the GBM T_1 signal, the GBM tumour burden signal and the healthy T_1 signal. The like-for-like comparison, between GBM T_1 and healthy T_1 , is ordered consistently across every discriminative descriptor: normalised high-frequency energy is ≈ 0.041 for GBM against ≈ 0.056 for healthy; the spectral centroid is ≈ 0.46 against ≈ 0.63 ; and the spectral entropy is ≈ 2.29 against ≈ 2.67 nats. The tumour burden signal extends the same ordering, with the lowest high-frequency energy (≈ 0.029) and the lowest centroid (≈ 0.41) of the three groups.

Two features of Figure 10 deserve comment. First, normalised low-frequency energy is ≈ 0.95 for all three groups and therefore carries essentially no discriminative information: with $E_{\text{low}} + E_{\text{high}} = 1$ and a spectrum overwhelmingly concentrated at low frequencies, E_{low} is saturated, and the informative contrast lives entirely in the small high-frequency residual. Second, the entropy ordering places the tumour burden signal (≈ 2.40) above the GBM T_1 signal (≈ 2.29) while both remain below healthy, so entropy and centroid do not induce the same ranking of the two GBM-derived signals; they measure dispersion and location of the spectrum respectively, and need not agree.

By Proposition 3.3, the centroid and entropy comparisons are unaffected by any global intensity gain between the cohorts, and the normalised energies inherit the same invariance. The ordering reported here is therefore not an artefact of differing intensity scales, although Section 5.4 explains why it is not thereby free of all acquisition effects.

The distributional comparisons behind these means are shown in Figures 11 and 12. Figure 11 restricts attention to the T_1 signal and therefore constitutes the methodologically cleanest contrast,

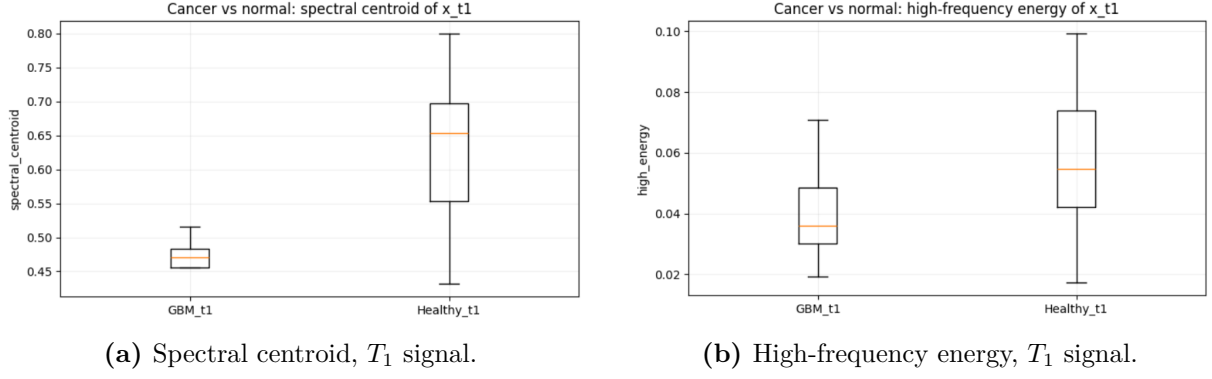


Figure 11. Like-for-like comparison of spectral descriptors computed from the T_1 signal in both cohorts. The healthy cohort exhibits both a higher spectral centroid and greater high-frequency energy, consistent with signal energy distributed over a broader range of graph harmonics.

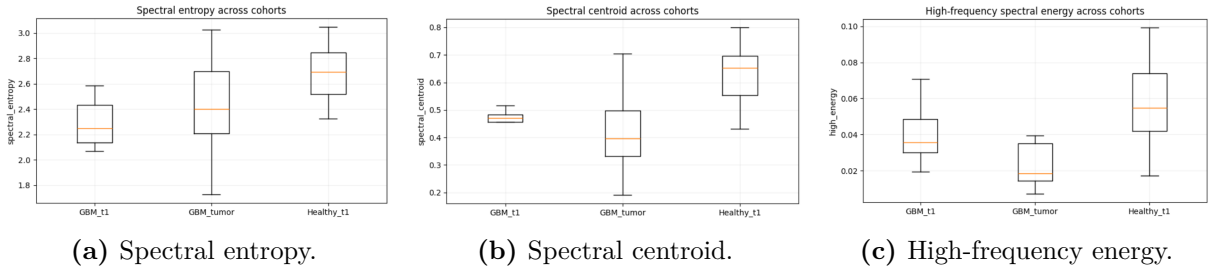


Figure 12. Distributions of the three discriminative spectral descriptors across all groups, including the tumour burden signal. The healthy cohort occupies the highest-entropy, highest-centroid and highest high-frequency region in each panel, while the GBM-derived signals are comparatively concentrated in smooth graph harmonics.

since both groups carry the same physical quantity; Figure 12 adds the tumour burden signal for reference.

4.3. Multiscale Wavelet Separation. Figure 13 reports multiscale graph wavelet energy at $\tau \in \{1, 5, 10, 20\}$. The separation is large at every scale but decreases in absolute magnitude as τ grows: healthy T_1 wavelet energy has median ≈ 30 at $\tau = 1$, ≈ 7 at $\tau = 5$, ≈ 2.5 at $\tau = 10$ and below 1 at $\tau = 20$, while the tumour burden signal remains below ≈ 0.5 throughout. This decay is exactly what Equation (3.18) predicts for the heat kernel: as τ increases, the multiplier $e^{-2\tau\lambda_k}$ suppresses every non-constant mode, so all signals converge towards their graph-mean component and the discriminative contrast migrates into the fine scales.

Fine-scale wavelet energy is the informative regime for a second reason. A signal with large energy at small τ necessarily retains localised irregularity on the graph, which by Proposition 3.2 corresponds to larger Dirichlet energy and hence to a larger smoothness penalty in the learning objective (1.1). The descriptive multiscale analysis and the regulariser are therefore measuring the same property of the signal at different points in the pipeline.

Figure 14 shows the corresponding node-wise coefficients for a representative GBM tumour signal, making the vertex-domain effect of increasing τ explicit: local peaks and abrupt node-to-node transitions present at $\tau = 1$ are progressively suppressed until only broad graph structure survives.

4.4. Geometric Deep Learning Classification. Figure 15 reports the training behaviour and held-out performance of the smoothness-regularised graph neural network over 150 epochs. Three patterns are visible.

- (a) The total objective decreases monotonically and almost identically on the training and test splits, from ≈ 0.68 to ≈ 0.11 and ≈ 0.09 respectively, with the test curve running slightly below the training curve throughout.

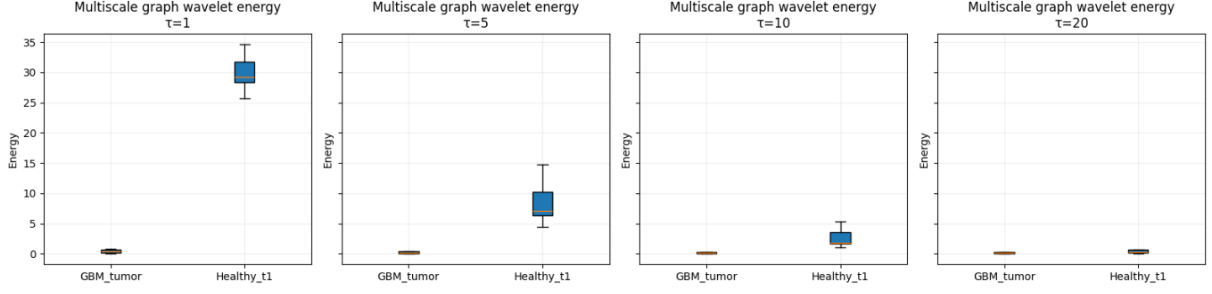


Figure 13. Multiscale graph wavelet energy across groups at scales $\tau = 1, 5, 10, 20$. Group separation is present at all scales but largest in absolute terms at $\tau = 1$, where healthy T_1 signals display a median energy of roughly 30 against below 0.5 for the GBM tumour signal.

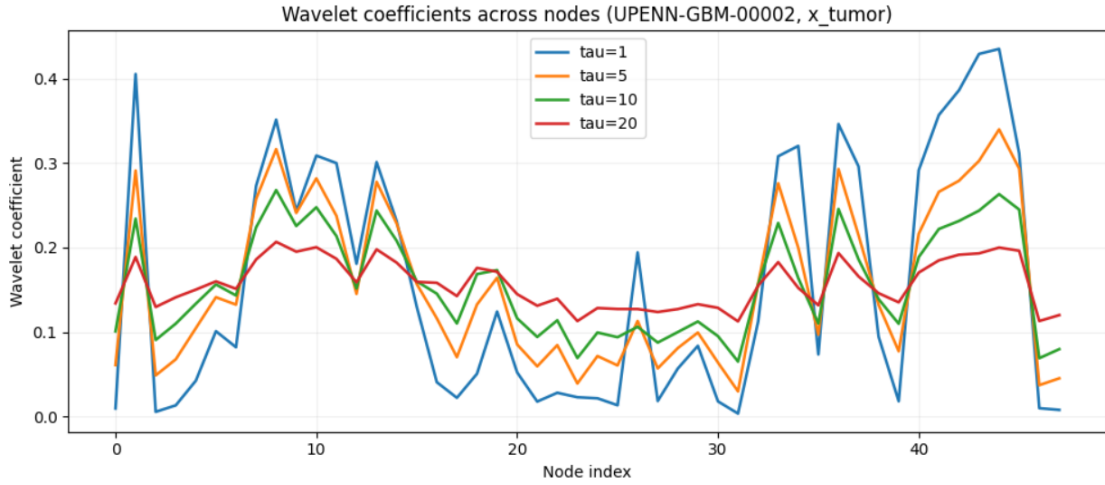


Figure 14. Wavelet coefficients across atlas nodes for a representative GBM tumour signal at scales $\tau = 1, 5, 10, 20$. Small scales retain local peaks and abrupt transitions; larger scales suppress these irregularities and expose broader graph structure.

- (b) Classification accuracy improves in discrete steps, as expected for small splits: training accuracy passes through 12/19, 14/19, 16/19 and 18/19 before reaching 19/19 near epoch 105, while test accuracy moves from 6/9 through 8/9 to 9/9 near epoch 62.
- (c) The prediction loss falls by more than an order of magnitude while the smoothness penalty *rises* from ≈ 5 to ≈ 37 .

The third pattern is the most informative, and it is not a contradiction. The optimiser minimises the *sum* in (1.1), so an increase in $\mathcal{L}_{\text{smooth}}$ is admissible whenever it buys a larger decrease in $\mathcal{L}_{\text{pred}}$. That the plotted total objective remains close to the prediction loss throughout, despite the penalty growing to ≈ 37 , indicates that λ is small in this configuration: the regulariser is shaping the embedding geometry without dominating the objective. The behaviour is the intended one. The penalty does not prohibit task-relevant graph variation; it prices it, so that node-to-node variation is purchased only where it earns discriminative value [44]. Read through Proposition 3.7, training is a competition between a task gradient that injects high-frequency structure into $H^{(L,s)}$ and a diffusion step that removes it, and the converged embedding is the fixed point of that competition.

The test confusion matrix shows 6/6 healthy and 3/3 GBM subjects correctly classified. The PCA projection of the learned subject embeddings (Figure 16) shows the corresponding geometry: the GBM subjects spread over a broad region spanning roughly -3.3 to -0.7 along the first principal component, whereas the healthy subjects collapse into an extremely tight cluster near $\text{PC}_1 \approx 1.15$ with negligible within-class spread.

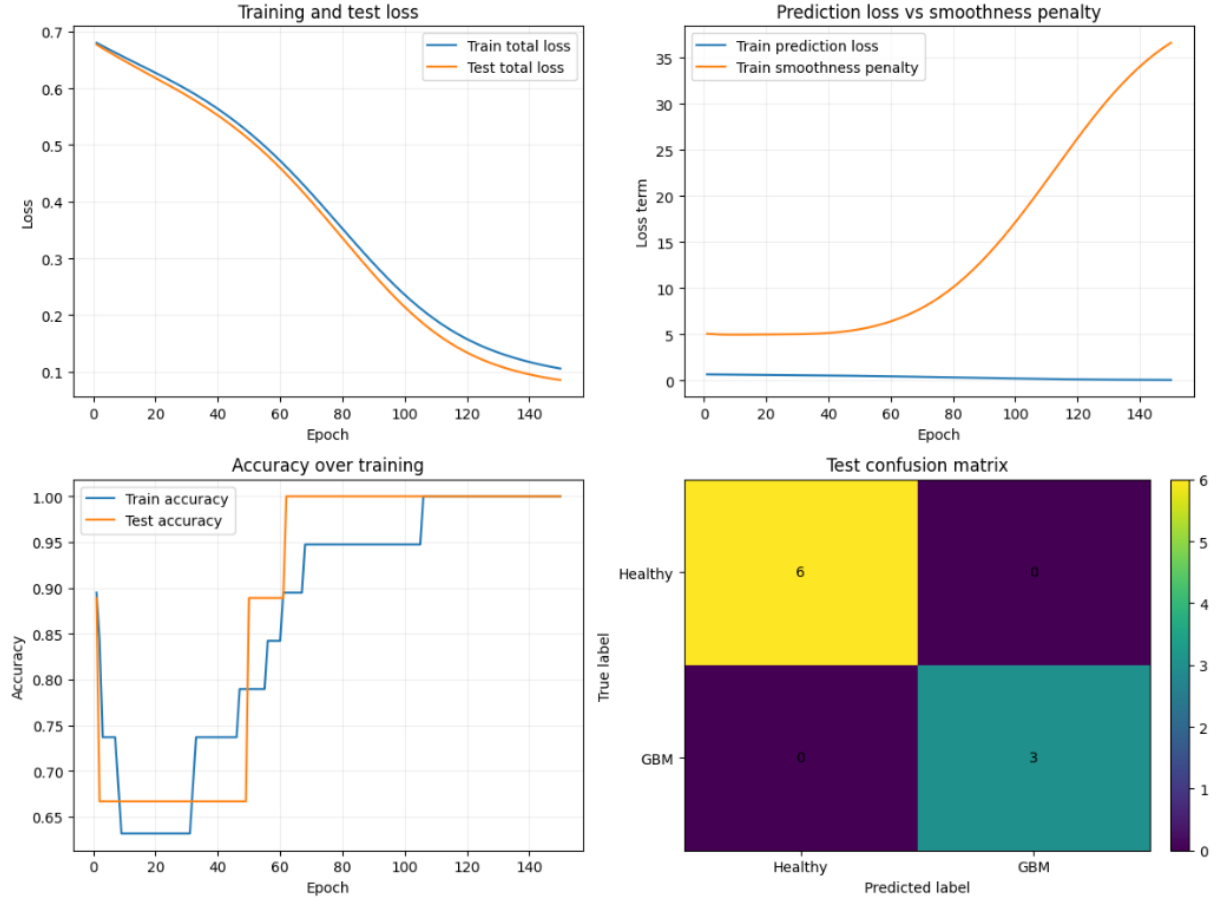


Figure 15. Geometric deep learning results over 150 training epochs. Top left: total objective on the training and test splits. Top right: the task-specific prediction loss against the Dirichlet smoothness penalty. Bottom left: training and test accuracy. Bottom right: the test confusion matrix, with all 6 healthy and all 3 GBM held-out subjects classified correctly.

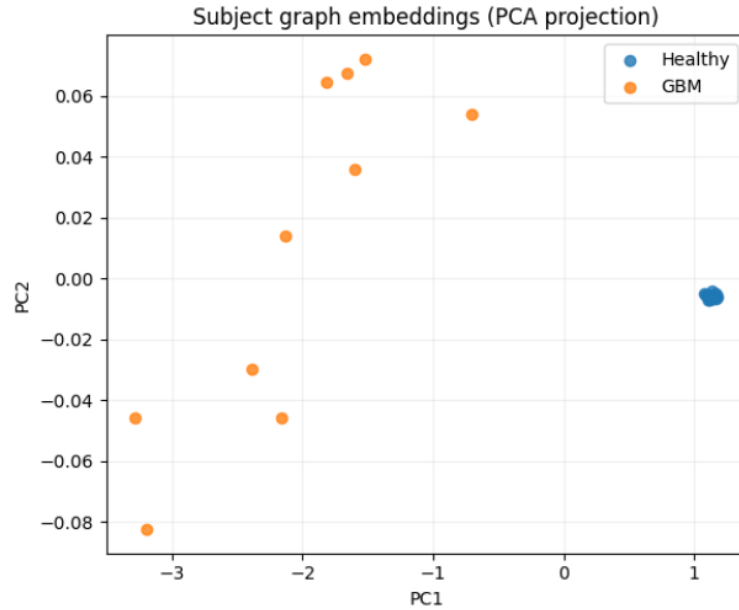


Figure 16. PCA projection of the learned subject-level graph embeddings. GBM subjects spread broadly along the first principal component while healthy subjects collapse to a near-degenerate cluster. The asymmetry is discussed in Section 5.4.

This asymmetry warrants caution rather than celebration. A class that collapses to a near-point in embedding space is the signature of a feature the network finds trivially and uniformly satisfied across that class. Combined with the fact that test accuracy saturates well before the training accuracy does—the model classifies all held-out subjects correctly while still misclassifying training subjects—the pattern is more consistent with an easily separable, globally consistent cohort difference than with a subtle pathological signature recovered from individual anatomy. Section 5.4 identifies the most plausible such difference.

5. DISCUSSION

5.1. Principal Findings. The descriptive analyses establish that the cohorts differ in how signal energy is distributed over graph frequencies and graph scales:

- GBM T_1 signals concentrate their spectral energy at lower graph frequencies ($\lambda \approx 0.23$) than healthy T_1 signals ($\lambda \approx 0.48$);
- healthy T_1 signals exhibit a higher spectral centroid, higher spectral entropy and greater normalised high-frequency energy;
- normalised low-frequency energy is saturated and non-discriminative, so the usable contrast lies in the high-frequency residual;
- multiscale wavelet energy separates the groups at every scale examined, most strongly in absolute terms at $\tau = 1$;
- disease effects are localised over specific atlas nodes and subnetworks rather than distributed uniformly.

Taken together these support the use of graph-structured regularisation: the graph carries biologically meaningful spatial organisation [1, 3], and the spectrum provides an interpretable basis for quantifying departures from typical organisation [17, 19, 20].

5.2. Relation to Prior Work. This framework joins two literatures that are usually pursued separately. GSP has been used to analyse functional and structural brain imaging by treating cortical measurements as signals more or less aligned with the anatomical graph [17, 18, 19, 21, 20]; geometric deep learning has produced powerful graph classifiers, including for neuroimaging cohorts [23, 25, 26, 27, 32, 33, 34]. The specific contribution here is to make the *same* spectral object—graph Dirichlet energy—serve as both the descriptive biomarker (through the GFT and spectral graph wavelets [8, 15]) and the training-time regulariser, which Propositions 3.6 and 3.7 identify with Laplacian-based semi-supervised smoothing [44, 45] and with heat diffusion on the graph [71]. The model’s inductive bias is thereby stated explicitly, measured in the same units as the descriptive analysis, and equipped with a stability condition, rather than being left implicit in the architecture.

5.3. Stability, Robustness and Denoising. The design pairs a fixed shared graph with subject-varying signals, which confers three stability properties:

- (1) *Common node semantics.* Every node denotes the same atlas region across subjects, because both cohorts are registered to MNI152 and parcellated with Harvard–Oxford [60, 61].
- (2) *Shared spectral basis.* A single eigenbasis U serves all GFT analyses, so spectral coefficients are directly comparable [11].
- (3) *Regularised learning.* The Dirichlet penalty limits sensitivity to noisy subject-specific node fluctuations [45].

The third property admits a precise statement. Suppose the observed signal is $x = x^* + \varepsilon$ with ε an additive perturbation. Since L is symmetric,

$$\mathcal{E}(x) = \mathcal{E}(x^*) + 2(x^*)^\top L\varepsilon + \mathcal{E}(\varepsilon), \quad \mathcal{E}(\varepsilon) = \sum_{k=1}^N \lambda_k |\hat{\varepsilon}_k|^2. \quad (5.1)$$

Noise concentrated at high graph frequencies therefore inflates the Dirichlet energy in proportion to λ_k , and by Proposition 3.7 the smoothing step attenuates precisely those modes most strongly. The penalty thus functions as a denoiser as well as a structural prior [8]. The cross term $2(x^*)^\top L\varepsilon$ vanishes in expectation when ε is zero-mean and independent of x^* , in which case

$\mathbb{E}[\mathcal{E}(x)] = \mathcal{E}(x^*) + \mathbb{E}[\mathcal{E}(\varepsilon)]$ and the penalty is inflated by a strictly positive, frequency-weighted noise term.

5.4. Limitations. The classification result must be read with its design in mind, and we state the constraints explicitly rather than in passing.

Cohort and acquisition confound. This is the principal limitation. The healthy and disease signals originate from two unrelated acquisition programmes: IXI was collected at three London hospitals, UPENN-GBM at the University of Pennsylvania, with different scanners, field strengths, sequences and reconstruction pipelines. Cohort membership is therefore perfectly confounded with site, and no analysis of this dataset can attribute the observed separation to pathology rather than to acquisition. The near-degenerate healthy cluster in Figure 16, and the saturation of test accuracy before training accuracy, are both consistent with the network exploiting a globally consistent, easily detected cohort-level difference. Proposition 3.3 removes one specific channel for this confound—a global intensity gain—but scanner differences are not confined to a global gain, and no invariance argument can neutralise them in general. Resolving this requires either healthy controls imaged under the UPENN-GBM protocol or explicit intensity harmonisation [70, 65] with a site-shuffled negative control.

Sample size and evaluation protocol. The analysed sample comprises 28 subjects with a single 19/9 split, so the reported accuracy is a point estimate on nine test subjects. The binomial confidence interval around 9/9 is wide, and a single split provides no estimate of variance across partitions. Small-sample neuroimaging results are known to be unstable and to overstate effect sizes [79, 80], and accuracy estimated without nested cross-validation is a recognised source of optimistic bias [81, 82]. Accordingly, the classification experiment should be read as a demonstration that the framework is implementable and internally consistent, not as evidence of a validated biomarker. Repeated stratified cross-validation with a permutation test for the null would be the minimum required to make a performance claim.

Heterogeneous signal types. The spectral comparison involves a continuous intensity signal and a binary-derived volume fraction. We have therefore foregrounded the like-for-like GBM T_1 versus healthy T_1 contrast (Figure 11) and treated the tumour signal as a descriptive reference, but conclusions that mix the two signal types remain weaker than those drawn within a type.

Fixed graph and limited modality. The shared graph fixes one topology for all subjects, which buys a common spectral basis at the cost of ignoring subject-specific connectivity [2, 4]. Only T_1 and the binary tumour mask are used, leaving the post-contrast T_1 , T_2 and FLAIR sequences and the graded sub-region labels of UPENN-GBM unexploited [41, 57].

5.5. Future Work. Four directions follow directly. First, and most importantly, the cohort confound should be addressed by sourcing healthy controls under a matched protocol or by harmonising intensities and verifying that separation does not survive label permutation. Second, the evaluation should move to repeated stratified cross-validation with permutation-based significance testing [81], and the cohort should be enlarged, for example by combining UPENN-GBM with the TCGA glioma collections [58, 59], to enable cross-site validation. Third, the fixed atlas graph could be replaced by subject-specific structural or functional connectomes, or learned from the data under smoothness assumptions [49, 50, 51], trading the shared spectral basis for subject specificity. Fourth, richer spectral filters—higher-order Chebyshev polynomials [26], attention-weighted aggregation [28] or anatomically adapted wavelets [22]—could sharpen both the descriptive biomarkers and the learned representations, and the stability bound (3.31) offers a principled starting point for selecting λ rather than tuning it blindly.

6. CONCLUSION

This paper develops a graph signal processing pipeline for atlas-based connectome analysis of glioblastoma multiforme, built on public, reproducible imaging resources [40, 41, 42]. Its contributions are a shared 48-node Harvard–Oxford graph in MNI152 space that gives both cohorts identical node semantics and a single spectral basis; graph signal definitions for T_1

intensity and tumour burden; spectral biomarkers together with a proof that the centroid and entropy are invariant to global rescaling while band energies are not; a multiscale graph wavelet analysis; a diffusion-theoretic account of the Dirichlet penalty, including its identification with an explicit Euler step of the graph heat equation and the resulting stability bound $\eta\lambda \leq 1/\lambda_N$; and a geometric deep learning classifier trained under that penalty.

The central claim is that connectome-aware analysis is valuable both descriptively and predictively, and that a single object serves both roles. Descriptively, the graph Fourier basis and graph wavelets localise where disease-related information sits across graph frequencies and scales [8, 15, 17]. Predictively, the same Laplacian supports the regularised objective $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \lambda\mathcal{L}_{\text{smooth}}$, which is principled because it encodes the assumption that neighbouring regions should carry related latent representations except where the task benefits from preserving edge-wise contrast [44, 45], and analysable because that penalty is a low-pass diffusion operator with a known admissible range for λ .

The empirical classification result is reported as a feasibility demonstration rather than as a validated biomarker: with 28 subjects drawn from two distinct acquisition programmes, cohort effects cannot be separated from disease effects, and resolving that confound is the necessary next step. The mathematical framework, by contrast, is independent of this particular dataset, and we expect the combination of spectral analysis, graph wavelets and smoothness-regularised geometric learning [23, 24] to remain a coherent basis for connectome-aware disease modelling as larger and better-matched cohorts become available.

DATA AND CODE AVAILABILITY

Both imaging resources used in this study are publicly available. The IXI dataset is distributed by the Biomedical Image Analysis Group, Imperial College London [40]. The UPENN-GBM collection is distributed by The Cancer Imaging Archive under its published data use policy [41, 42, 43]. The MNI152 template [60] and the Harvard–Oxford atlas [61] are distributed with standard neuroimaging toolkits [66, 73]. No new human-subject data were collected for this study, and all analyses were performed on de-identified public data.

ACKNOWLEDGEMENTS

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COMPETING INTERESTS

The authors declare no competing interests.

Table 3. Summary of notation.

Symbol	Meaning
$G = (V, E, W)$	Weighted undirected connectome graph with node set V , edge set E and adjacency matrix W .
N	Number of graph nodes (here $N = 48$).
D	Degree matrix, $D_{ii} = \sum_j W_{ij}$.
L	Combinatorial graph Laplacian, $L = D - W$.
L_{sym}	Symmetric normalised Laplacian, $I - D^{-1/2}WD^{-1/2}$.
U, Λ	Laplacian eigenvectors and eigenvalues, $L = U\Lambda U^\top$.
λ_k	k -th Laplacian eigenvalue, $0 = \lambda_1 \leq \dots \leq \lambda_N$.
$x^{(s)}$	Graph signal for subject s .
$\hat{x}$	Graph Fourier transform of x , $\hat{x} = U^\top x$.
$\mathcal{E}(x)$	Dirichlet energy, $x^\top Lx$.
$E_{\text{low}}, E_{\text{high}}$	Low- and high-frequency spectral energy.
C_{spec}	Spectral centroid, Equation (3.13).
H_{spec}	Spectral entropy, Equation (3.14).
g, τ	Spectral wavelet kernel and wavelet scale.
$H^{(L,s)}$	Final-layer node embedding matrix for subject s .
$\mathcal{L}_{\text{pred}}$	Prediction loss (binary cross-entropy).
$\mathcal{L}_{\text{smooth}}$	Dirichlet smoothness penalty on learned embeddings.
λ	Regularisation weight balancing prediction and smoothness.
η	Optimiser learning rate.

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DEPARTMENT OF SCIENCE AND TECHNOLOGY, SCIENTIFIC RESEARCHERS LLC, INDIANAPOLIS, INDIANA, USA
 Email address: asreera110@scientificresearchers.org

DEPARTMENT OF SCIENCE AND TECHNOLOGY, SCIENTIFIC RESEARCHERS LLC, INDIANAPOLIS, INDIANA, USA
 Email address: anushadronamraju20@scientificresearchers.org

DEPARTMENT OF SCIENCE AND TECHNOLOGY, SCIENTIFIC RESEARCHERS LLC, INDIANAPOLIS, INDIANA, USA
 Email address: satyadronamraju11@scientificresearchers.org